Programming Project 4 - Implementation plans

vsfs_link:

Operations Requried

1. Ensure source path exist
2. Find the correct inode for source path
3. check that the file is not a link
4. look into the directory holding desination path and find which data block it is in
5. find a empty directory entry in said data block
6. check that there is no name conflict
7. update new directory entry
8. increment link-count
9. write changse to disk

vsfs_unlink:

Operations Requried

1. Ensure source and destination paths exist
2. Find the correct inode for source path
3. find the directory the link is in
4. Verify that the file is actually a link
5. remove directory entry
6. reduce inode link-count
7. – if link-count > 0 – change inode
8. – if link-count $= 0$ – read the inode free blocks that are referenced free indirect block if it exist re-write the inode clear the allocation bit mark inode as free - reset metadata
9. write changes to disk